

GABRIEL POPA

Senior AI Full Stack Engineer

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ABOUT ME

Senior AI Full Stack Engineer with 10+ years of experience delivering production-grade platforms across the **software, SaaS, enterprise workflow, and data automation industry**, combining **React 16–18, TypeScript 4–5, Node.js 14–20, Python 3.8–3.12, FastAPI 0.9+, Next.js 12–14, LLM integrations, RAG pipelines**, and cloud-native backend systems to build secure, scalable, and user-facing AI products. Strong background in designing **REST APIs**, modern frontend architecture, prompt-driven workflows, vector search experiences, and resilient microservices, while fixing real production bugs, reducing latency, improving model response quality, and modernizing legacy applications into maintainable AI-enabled platforms used by internal teams and external customers.

SKILLS

- **Frontend: React 16–18, Next.js 12–14, TypeScript 4–5, JavaScript ES6+, HTML5, CSS3, Tailwind CSS, Redux Toolkit, Context API, Material UI, Responsive UI, SPA architecture**
- **Backend & APIs: Node.js 14–20, Express.js 4+, NestJS 9–10, Python 3.8–3.12, FastAPI 0.9+, Django 3–4, REST API design, GraphQL, WebSocket services, Microservices, Authentication, JWT, RBAC**
- **AI / LLM / Search: OpenAI API, Anthropic API, RAG, Prompt Engineering, Function Calling, AI Agents, LangChain 0.1+, LlamaIndex 0.9+, Vector Search, Semantic Search, Embeddings, Pinecone, Weaviate, FAISS, Elasticsearch, OpenSearch**
- **Data & Messaging: PostgreSQL 12–16, MySQL 5.7–8, MongoDB 4–7, Redis 6–7, Kafka, RabbitMQ, AWS SQS/SNS**
- **Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Terraform, GitHub Actions, GitLab CI/CD, Nginx, Linux, Serverless deployments**
- **Observability / Quality / Security: Jest, Playwright, Cypress, Pytest, Postman, OpenTelemetry, Grafana, Prometheus, CloudWatch, Sentry, OAuth2, Rate limiting, Audit logging**

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Soft Build | Bucharest, Romania (*Jan 2024 – May 2026*)

- Led delivery of an **AI-assisted enterprise workflow platform** using **React 18, TypeScript 5, Node.js 20, and Python 3.11**, giving business users a unified workspace for document search, task automation, and conversational assistance across internal knowledge sources.
- Designed a **RAG-based retrieval layer** with **OpenAI, LangChain, PostgreSQL**, and **vector indexing**, which improved answer relevance by **31%** after restructuring chunking strategy, metadata filtering, and prompt templates for multi-step queries.
- Built dynamic frontend modules in **Next.js 14** and **Tailwind CSS** for chat, citation panels, prompt history, and admin controls, improving user workflow completion by **27%** through clearer state handling and faster interaction patterns.
- Implemented secure backend services with **NestJS, FastAPI, JWT**, and **RBAC**, enabling controlled access to AI tools, knowledge sources, and audit records in a multi-role enterprise environment.
- Resolved a recurring production issue where long AI responses broke client rendering and caused intermittent timeout failures, then introduced response streaming, retry guards, and chunk-safe rendering that reduced related incidents by **36%**.
- Migrated older synchronous API flows into asynchronous worker-based pipelines with **Redis** and **RabbitMQ**, which stabilized document ingestion, improved retry behavior, and reduced failed processing jobs by **24%**.
- Created prompt evaluation utilities and regression checks so new prompt or model changes could be validated before release, reducing quality drift and cutting manual review effort by **22%** for the delivery team.
- Improved slow search endpoints by refactoring SQL queries, adding cache-aware retrieval, and tuning database indexes, which lowered average response time by **33%** during peak internal usage.

- Worked closely with product and design stakeholders to implement new AI features such as summarization, follow-up suggestions, and answer source tracing, making the platform more explainable and increasing adoption across non-technical teams.
- Strengthened release quality by expanding **Jest**, **Playwright**, and API contract coverage for critical user journeys, preventing regressions in prompt execution, auth flows, and document upload behavior.
- Added observability with **OpenTelemetry**, **Grafana**, and structured logging, which made it easier to trace model latency, retrieval failures, and third-party API issues during production incidents.
- Supported cloud deployment and environment hardening with **Docker**, **Kubernetes**, and **AWS**, ensuring scalable rollout of AI services while keeping configuration, secrets, and background jobs consistent across environments.

Software Engineer

Next Apps | Ghent, Belgium (Nov2020 – Nov 2023)

- Built and enhanced customer-facing **full stack SaaS applications** using **React 17**, **TypeScript 4**, **Node.js 16**, and **PostgreSQL**, with a focus on intelligent workflow features, dashboard usability, and backend reliability.
- Implemented early **AI-driven assistance features** for search, text classification, and content recommendations using **Python**, API-based LLM services, and structured business rules to improve user productivity in data-heavy screens.
- Developed reusable frontend components and state models that simplified complex data entry and review workflows, increasing successful form completion by **26%** and reducing user-side correction loops across several modules.
- Solved a difficult frontend synchronization bug where stale cached data caused inconsistent status changes between pages, then introduced normalized state updates and better invalidation logic that reduced duplicate support tickets by **21%**.
- Expanded backend integrations with third-party APIs and internal services through **Express.js** and event-based processing, improving synchronization accuracy and reducing delayed background updates by **18%**.
- Migrated legacy UI sections from older React class-based patterns into modern functional components with hooks, making the codebase easier to test, easier to extend, and significantly cleaner for future AI-driven features.
- Improved reporting and analytics workflows by redesigning API payloads, optimizing joins, and introducing pagination and lazy-loading patterns that reduced heavy dashboard load times by **29%**.
- Added AI-enabled search refinement using semantic ranking and metadata filtering so users could find records faster, which improved search success behavior and reduced repeated query attempts during complex lookup tasks.
- Built internal tooling for monitoring failed jobs, API retries, and sync mismatches, helping support teams identify issue patterns faster and shortening resolution time during production incidents.
- Strengthened CI/CD and test pipelines with **GitLab CI**, **Jest**, and **Cypress**, leading to more consistent deployments and fewer regressions in high-traffic product workflows.
- Collaborated in architecture discussions for modular services and shared component design, helping prepare the platform for later migration toward more scalable AI-enabled services and modern cloud deployment practices.

Software Developer

Datamart | Stockholm, Sweden (Feb 2016 – Sep 2020)

- Developed enterprise web applications and internal productivity tools using **JavaScript**, **React 16**, **Node.js 12–14**, **Python 3**, and **MySQL**, supporting business operations with stable dashboards, workflow automation, and data processing modules.
- Built backend services for document handling, reporting, and data transformation, then extended them with early machine-learning-assisted classification and recommendation flows for internal efficiency use cases.
- Improved data import pipelines that were failing on malformed records and inconsistent schema mapping, then added stronger validation, retry handling, and logging that reduced import-related processing errors by **28%**.
- Created frontend views for analytics, search, and record management that made large datasets easier to navigate, helping teams complete operational review tasks faster and with fewer manual spreadsheet exports.
- Resolved repeated API bottlenecks caused by oversized payloads and blocking processing steps, then introduced batching and lighter response contracts that improved service throughput by **23%**.
- Worked on migration of older monolithic modules into cleaner service-oriented backend layers, allowing teams to release features more safely and isolate failures without affecting the entire application.
- Added search enhancements with **Elasticsearch** for record discovery and document lookup, increasing search speed and improving relevance for high-volume operational queries.

- Implemented role-aware access control, audit trails, and safer input validation in critical screens, making the system more secure and more compliant for business users handling sensitive records.
- Partnered with QA and business stakeholders to reproduce hard-to-track issues in production data flows, then delivered targeted fixes that improved reliability and restored trust in several internal workflows.
- Contributed to deployment automation with **Docker** and script-based pipelines, helping standardize environments and reduce machine-specific inconsistencies during release preparation.

Junior Software Developer

Berg Software | Timișoara, Romania (Sep2014 – Mar 2016)

- Supported development of business web applications using widely adopted technologies of the period, including **JavaScript, jQuery, AngularJS, PHP, Node.js, HTML, CSS, and MySQL**, while learning production delivery practices in a collaborative engineering team.
- Built UI enhancements, validation logic, and backend-connected forms for internal and client-facing applications, improving day-to-day usability and reducing repeated user entry mistakes across operational screens.
- Fixed recurring bugs related to form submission, browser inconsistency, and incomplete server responses, then helped stabilize fragile workflows that had been generating repeated manual correction work.
- Assisted with database queries, stored procedures, and reporting features that improved operational visibility and reduced turnaround time for common support and reporting requests by **17%**.
- Worked on early API integrations and backend maintenance tasks, learning how to trace data issues across frontend, server, and database layers in real business systems.
- Improved old interface sections by simplifying DOM-heavy logic and removing duplicated client-side behavior, making pages easier to maintain and reducing front-end rendering issues by **19%**.
- Helped troubleshoot release issues caused by environment mismatches and configuration drift, then supported cleanup of deployment steps so builds were more predictable across staging and production.
- Participated in code reviews, bug verification, and feature refinement with senior engineers, which accelerated learning while also improving implementation quality and testing discipline.
- Contributed to gradual modernization of selected modules so older pages were easier to extend later, providing a stronger base for future migration toward more maintainable full stack application patterns.

EDUCATION

Bachelor's Degree in Computer Science

University of Bucharest | (2010 – 2014)

- Focused on practical software engineering fundamentals that translated directly into building reliable web systems and data-driven applications.

EDUCATION

- **AWS Certified Cloud Practitioner**
- **Microsoft Certified: Azure Fundamentals**
- **OpenAI API Developer Certificate**
- **Docker & Kubernetes Foundations Certificate**
- **Postman API Fundamentals Student Expert**